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HW_5_Extra Credit Completed Item List

**** This submission is intentionally partially completed and should only be assessed as extra credit. I completed homework 4 as my primary submission and then began to work on this in my spare time knowing I would not have enough time to fully complete it.****

((20 points) Update the chicken domain to implement the modified rules, using 4 coops, 64 chickens and 256 tournaments.)

- Updated input variables so that T = 256 istead of T = 1000
- Added the "sack" variable to chickens
- Updated the "CountCrowns()" function to update the sack component after each tournament
- Created the ShareObservations() and tested that it functions correctly

```
Tournament: 1
chicken 0 memory of chicken 63 before share
[ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  1.  1.  1. -1.  0.]
chicken 63 memory of chicken 63 before share
[ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  1.  1.  1. -1.  0.]
chicken 0 memory of chicken 63 after share
[ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  1.  1.  1. -1.  0.]
```

- Created transfercrowns() and tested that it functions correctly

```
Battle zone count: 40
Tournament: 4
chicken 0 crowns
0
chicken 55 crowns
5
transfer 1 crown
chicken 0 crowns
1
chicken 55 crowns
4
```

- Added sack_owner as a chicken class component to track which chicken are sharing a sack
- Created ExtendSackOwnership() and tested that it functions correctly

```
def ExtendSackOwnership(self, Chick_j_ID, ent_set):
    # assign the new owners
    new_owners_i = list(self.sack_owner) + [Chick_j_ID]
    new_owners_j = list(ent_set.chickens[Chick_j_ID].sack_owner) + [self.index]


    # update chicken components
    updated_chicken = replace(self, sack_owner = new_owners_i)
    updated_j = replace(ent_set.chickens[Chick_j_ID], sack_owner = new_owners_j)

    # udpate global
    chick_list = list(ent_set.chickens)
    chick_list[self.index] = updated_chicken
    chick_list[Chick_j_ID] = updated_j

    return replace(ent_set, chickens=tuple(chick_list))
```

- Created the function, ShareCount() to consider chickens that have teamed up and tally their crown totals accordingly by adding up all their crowns and diving by the number of chickens on the team
    - 0 and 55 are sharing:
        - Before ShareCount()
            -
```
Tournament: 9
[(chick) (# of crowns)]
[[ 0.  0.]
 [ 1.  5.]
```
            -
```
[53.  0.]
[54.  1.]
[55.  8.]
[56.  0.]
```
        - After ShareCount()
            -
```
 [63.  1.]]
[(chick) (# of crowns)]
[[ 0.  4.]
 [ 1.  5.]
 [ 2.  0.]
```
            -
```
[53.  0.]
[54.  1.]
[55.  4.]
[56.  0.]
```

(10 points) Implement NPCs (Non-player chickens) that execute a simple strategy of moving to the coop where they have highest ability. These chickens do not use the new actions. There will be 56 NPCs.